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# QUESTION 14: Effect of Image Scaling on Feature Detection
# Description: Loads an image from file and resizes it
#              to analyze feature count variations at multiple scales.
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt

# 1. Load Input Image
# *** MAKE SURE TO DOWNLOAD THE ATTACHED IMAGE AND SAVE IT AS 'input_image.png' ***
input_path = 'SAM.jpg'
loaded_img = cv2.imread(input_path)

if loaded_img is None:
    print(f"Error: Could not load image from '{input_path}'. Please ensure the file exists.")
    exit()

# Convert the loaded image to grayscale for processing
img_14 = cv2.cvtColor(loaded_img, cv2.COLOR_BGR2GRAY)

# 2. Define scales to test
scales = [1.0, 0.75, 0.5, 0.25]
orb = cv2.ORB_create(nfeatures=500)

plt.figure(figsize=(16, 4))
print("--- Scale Analysis Log ---")

# 3. Downscale image and run ORB feature detection on each scale
for idx, scale in enumerate(scales):
    new_w = int(img_14.shape[1] * scale)
    new_h = int(img_14.shape[0] * scale)
    scaled_img = cv2.resize(img_14, (new_w, new_h), interpolation=cv2.INTER_LINEAR)

    keypoints = orb.detect(scaled_img, None)
    # The drawKeypoints function should use the grayscale image, not the original loaded_img.
    img_kp = cv2.drawKeypoints(scaled_img, keypoints, None, color=(0, 255, 0))

    print(f"Scale: {scale:4.2f}x | Resolution: {new_w}x{new_h} | Detected Keypoints: {len(keypoints)}")

    plt.subplot(1, len(scales), idx + 1)
    # The color conversion is necessary because matplotlib expects RGB, while OpenCV uses BGR (or loads as gray).
    # Since img_kp is BGR, conversion is correct.
    plt.imshow(cv2.cvtColor(img_kp, cv2.COLOR_BGR2RGB))
    plt.title(f"Scale: {scale}x\n({len(keypoints)} Features)", fontsize=10)
    plt.axis('off')

plt.tight_layout()
plt.show()
```

--- Scale Analysis Log ---

Scale: 1.00x | Resolution: 4928x3264 | Detected Keypoints: 500

Scale: 0.75x | Resolution: 3696x2448 | Detected Keypoints: 500

Scale: 0.50x | Resolution: 2464x1632 | Detected Keypoints: 500

Scale: 0.25x | Resolution: 1232x816 | Detected Keypoints: 500

Scale: 1.0x
(500 Features)



Scale: 0.75x
(500 Features)



Scale: 0.5x
(500 Features)



Scale: 0.25x
(500 Features)



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